

- 5 The curve  $C$  has polar equation  $r = a \cot\left(\frac{1}{3}\pi - \theta\right)$ , where  $a$  is a positive constant and  $0 \leq \theta \leq \frac{1}{6}\pi$ .

It is given that the greatest distance of a point on  $C$  from the pole is  $2\sqrt{3}$ .

- (a) Sketch  $C$  and show that  $a = 2$ .

[3]

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- (b) Find the exact value of the area of the region bounded by  $C$ , the initial line and the half-line  $\theta = \frac{1}{6}\pi$ . [4]

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